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Tunable cobalt-catalysed hydrogenation of allenes enabled by multiple metal–ligand cooperative functionalities

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Catalytic hydrogenation is essential in synthetic chemistry, with ongoing innovations aimed at enhancing the selectivity and efficiency of these reactions. The hydrogenation of multisubstituted allenes, however, presents a long-standing challenge due to the difficulty of controlling multiple selectivity factors simultaneously. Here we introduce a series of chiral pincer cobalt catalysts featuring multiple metal–ligand cooperative functionalities. These catalysts feature an 'N–H' moiety as an outer-sphere binding site and an N-heterocycle group as a hemilabile basic site, enabling the use of structurally diverse tridentate ligands for selective hydrogenation of functionalized allenes. This design liberates a coordination site for H_2 activation and enhances selectivity control through the structural tuning of the N-heterocycle group. The catalysts exhibit exceptional chemo-, regio-, enantio- and *Z*/*E*-selectivities, along with broad functional group tolerance, enabling access to all possible semihydrogenation products of multisubstituted allenes. Mechanistic studies uncover a distinctive redox-neutral Co(I) catalytic cycle that facilitates heterolytic cleavage of H_2, assisted by the basic N-heterocycle on the ligand.

Catalytic hydrogenation reactions are crucial in synthetic chemistry and a variety of industrial processes[1]. A landmark achievement in homogeneous catalysis during the 1960s was the development of the $RhCl(PPh_3)_3$ complex, known as Wilkinson's catalyst, which profoundly advanced the use of catalytic hydrogenation in organic synthesis and the production of fine chemicals[2]. This innovation has spurred continuous advancements in designing molecularly defined transition-metal catalysts capable of hydrogenating diverse unsaturated compounds. Despite these advancements, achieving precise control over the product distribution of hydrogenation reactions, particularly for substrates with multiple unsaturated bonds, remains an important yet challenging objective[3–6]. This challenge is especially pronounced in the hydrogenation of allenes[7]. Allenes, characterized by their unique orthogonal cumulative π-system, are readily accessible and serve as versatile building blocks for numerous chemical transformations[8–14]. However, asymmetric semihydrogenation of multisubstituted allenes is particularly challenging due to the need for simultaneous control over chemo-, regio-, enantio- and *Z*/*E*-selectivities across competing reaction

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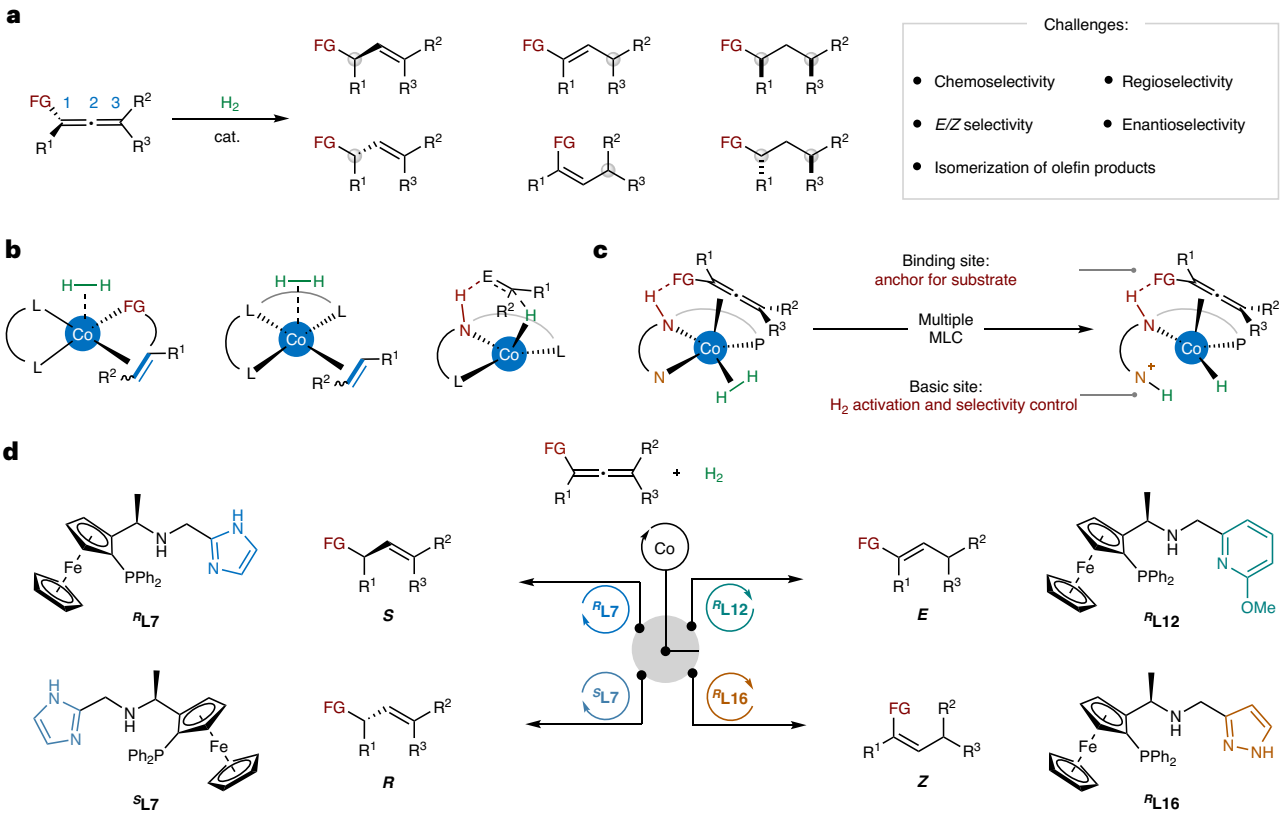


Fig. 1 | Divergent and selective hydrogenation of multisubstituted allenes. **a**, Long-standing challenges in selective hydrogenation of allenes. Potential challenges include chemoselectivity, regioselectivity, *E*/*Z*-selectivity and enantioselectivity, as well as undesired isomerization of olefin products. **b**, Three known catalytic models of cobalt-catalysed hydrogenation of different substrates. The pincer cobalt catalysts have been used primarily in the hydrogenation of non-functionalized alkenes or alkynes, while asymmetric hydrogenation of functionalized alkenes is more commonly achieved using cobalt catalysts supported by bidentate ligands. The MLC strategy is widely utilized in the catalytic hydrogenation of polar unsaturated compounds. **c**, Design plan: tunable hydrogenation of functionalized allenes enabled by pincer cobalt catalysts with multiple MLC functionalities. The first MLC functionality binds to the functionalized allene substrate, freeing a coordination site for H_2 activation, while the second MLC functionality facilitates H_2 activation and governs selectivity control. **d**, Our work: regio- and stereodivergent hydrogenation of multisubstituted functionalized allenes via ligand control. FG, functional group. For tetra-substituted allenes, $R^2 = R^3$; for racemic trisubstituted allenes, either R^2 or $R^3 = H$.

pathways (Fig. 1a). Compounding this difficulty, isomerization of semihydrogenation products and over-reduction to fully hydrogenated side products often diminish selectivity, underscoring the complexity of this transformation. Significant efforts have been devoted to addressing this issue, yet notable progress has been achieved primarily by the groups of Ma[15], Zhang[16] and Dong[17], each achieving excellent yields and selectivity for one of the semihydrogenation products. However, a regio- and stereodivergent hydrogenation of allenes that enables access to all possible semihydrogenation products has not been accomplished. Achieving this milestone would enable the efficient creation of libraries of stereodefined multisubstituted alkenes, which are in high demand for drug discovery and diversity-oriented synthesis[18–20].

Cobalt, an earth-abundant transition metal, has demonstrated remarkable potential in the development of new catalysts for asymmetric hydrogenation of various unsaturated compounds[21–26], including both alkenes[27–38] and polar unsaturated compounds such as ketones[39,40] and imines[41] (Fig. 1b). Among these achievements, pincer cobalt catalysts stand out due to their unique tridentate coordination mode, which provides robust stability and allows for more tunable steric and electronic properties compared to cobalt catalysts supported by bidentate ligands[42,43]. These advantages position pincer cobalt catalysts as promising candidates for achieving ligand-controlled regio- and stereodivergent semihydrogenation of multisubstituted allenes. Despite their structural advantages, pincer cobalt catalysts have been applied primarily to the hydrogenation of non-functionalized alkenes or alkynes[27,35], while functionalized alkenes are more commonly addressed using bidentate ligands[29,34,37]. We hypothesize that this discrepancy may be due to the absence of available coordination sites for H_2 activation when functionalized alkene substrates occupy two sites alongside a tridentate ligand[44]. Inspired by notable progress in the catalytic hydrogenation of polar unsaturated compounds through the metal–ligand cooperation (MLC) strategy[45–52], we propose the development of novel pincer cobalt catalysts. These catalysts would feature multiple MLC functionalities, combining the structural diversity of tridentate ligands for precise selectivity control with adequate coordination sites for the hydrogenation of functionalized non-polar unsaturated substrates, such as multisubstituted allenes. As shown in Fig. 1c, our pincer ligands integrate an 'N–H' moiety as a secondary-sphere binding site, enabling hydrogen-bonding interactions with the polar functional groups of allene substrate and liberating a coordination site for H_2 activation. Concurrently, an N-coordinating group, predominantly N-heterocycles, acts as a basic site, facilitating the heterolytic cleavage of H_2 in cooperation with cobalt centre via a distinctive redox-neutral Co(I) catalytic cycle[53]. This design not only promotes efficient H_2 activation but also allows for fine-tuned selectivity through structural modifications of the N-heterocycles. Using this approach, we demonstrate that four chiral *NNP*-pincer ligands, derived from readily available Ugi's amine scaffold, achieve precise selectivity control. This enables the regio- and stereodivergent hydrogenation of allenes, providing straightforward access to all possible multisubstituted alkene products (Fig. 1d).

Results and discussion

Reaction development

Our investigation began with cobalt-catalysed hydrogenation of multisubstituted allenyl phosphonate **1a**, aiming to optimize the reaction conditions. Initially, we tested various commercially available bidentate ligands (**L1**–**L6**) commonly used in cobalt-catalysed asymmetric hydrogenation[28,29]. However, these trials yielded unsatisfactory outcomes (Fig. 2a and Supplementary Table 3), confirming our concerns about low reactivity and selectivity in existing catalytic systems for this challenging transformation. These findings underscored the need for innovative catalyst development.

In line with our interest in cobalt hydride catalysis[25,54–56], we developed a library of multifunctional MLC pincer ligands, each based on the same chiral scaffold but featuring different heterocyclic units (**^{R}L7**–**^{R}L16**), and explored a structure–reactivity/selectivity relationship of these ligands in the hydrogenation of **1a** (Fig. 2b). The *NNP*-ligand **^{R}L7**, containing an imidazole group, enabled the synthesis of 1,2-hydrogenation product **2a** with 75% yield and high enantioselectivity, although regioselectivity remained moderate (95% enantiomeric excess (e.e.) and 3:1 regioselective ratio (r.r.)). Substituting imidazole with other heterocycles such as benzimidazole (**^{R}L8**), 4,5-diphenyl-imidazole (**^{R}L9**), pyridine (**^{R}L10**) and various substituted pyridines (**^{R}L11**–**^{R}L13**) generally led to reduced catalytic activity and selectivity for allyl phosphonate **2a**. Notably, **^{R}L12** with a methoxy-substituted pyridyl group reversed the regioselectivity, producing the 2,3-hydrogenation product **3a** with an excellent *E*-selectivity (78% yield, *E*/*Z* = 16:1, 1:5 r.r.). These results collectively demonstrate that the heterocycle moiety of ligand plays a crucial role in controlling regioselectivity, underscoring the need for further investigation into other heterocycle-based pincer ligands. However, introducing furan (**^{R}L14**) and thiophene (**^{R}L15**) resulted in poor efficiency and selectivity, while pyrazole-containing **^{R}L16** altered the stereoselectivity, favouring the alkenyl phosphonate (*Z*)-**3a** with notable regio- and stereoselectivity, although yield remained moderate (34% yield, *Z*/*E* = 49:1, 1:15 r.r.). Optimizing the reaction with 4 mol% of cobalt catalyst for 40 hours significantly enhanced the efficiency and selectivity (92% yield, *Z*/*E* = 49:1, 1:16 r.r.), establishing these as optimal conditions for *Z*-selective 2,3-hydrogenation. Further optimization for asymmetric 1,2-hydrogenation led to the best conditions using $Co(BF_4)_2{\cdot}6H_2O$ (2 mol%), **^{R}L7** (2.2 mol%) and zinc dust (5 mol%) in DMAc under 30 bar H_2, yielding (*S*)-**2a** with 93% yield, 95% e.e. and 14:1 r.r. As expected, using the opposite enantiomer of optimized ligand **^{S}L7** produced the *R* isomer of **2a**. Collectively, these findings establish a robust synthetic platform for tunable hydrogenation of tetra-substituted allenes, achieving comprehensive semihydrogenation products with excellent yield and chemo-, regio- and stereoselectivity through ligand control (Fig. 2c).

In our catalyst design, the N–H moiety on the ligand is critical for freeing a coordination site for H_2 activation through MLC. To validate this strategy, we synthesized and evaluated N-methylated variants of our ligands (**^{R}L17**–**^{R}L21**) in model reaction (Fig. 2b). Using the N-methylated imidazole ligand **^{R}L17** showed no impact on reaction efficiency or selectivity. However, no conversion occurred with ligands **^{R}L18**, **^{R}L19** and **^{R}L21**, indicating that the central N–H group is essential for substrate binding in the second coordination sphere. Interestingly, the ligand **^{R}L20**, featuring an N-methylated pyrazole, facilitated smooth reactivity but altered the stereoselectivity of alkenyl product compared with its N–H counterpart, **^{R}L16**. These findings indicate the critical role of N–H group in pyrazole for achieving *Z*-selective 2,3-hydrogenation.

The aforementioned results indicated that the heterocyclic group of ligand plays a critical role in selectivity control, thereby underscoring the necessity for further investigation into the structures of these catalysts. We synthesized corresponding Co(II) complexes by treating $CoCl_2$ with ligand in tetrahydrofuran, and solid-state structures of several complexes were determined by single-crystal X-ray diffraction. It was found that **^{R}L7-CoCl$_2$** and **^{R}L17-CoCl$_2$** exhibit a distorted-trigonal-bipyramidal geometry in tridentate coordination structures, whereas **^{R}L11**, **^{R}L12**, **^{R}L14** and **^{R}L15-CoCl$_2$** are bidentate coordination structures (Extended Data Fig. 1a and Supplementary Figs. 1–6). The dissociation free energies of N-heterocyclic ligands were further investigated, revealing that ligands **^{R}L7** and **^{R}L16** exhibit a strong preference for tridentate coordination (>5 kcal mol^{-1}), while **^{R}L12** and **^{R}L13-CoCl$_2$** exhibit significantly lower dissociation free energy of heterocyclic group (0.6 and 2.7 kcal mol^{-1}, respectively), implying an equilibrium between coordinated and dissociated states (Extended Data Fig. 1b and Supplementary Fig. 34). The structural difference among these cobalt catalysts may arise from the significantly distinct steric and electronic properties of heterocyclic groups.

Reaction scope

After establishing optimal reaction conditions, we explored the scope of 1,2-hydrogenation of tetra-substituted allenes **1** for synthesizing chiral allylic products **2**, as shown in Table 1. We were pleased to discover that various substituents on the allenyl phosphonates, including both alkyl and cycloalkyl groups, underwent successful transformations, yielding the desired products with high efficiency (**2a**–**2h**, up to 95% yield, 99% e.e. and 22:1 r.r.). The presence of electron-donating or electron-withdrawing groups on the aromatic ring did not significantly impact the reaction, consistently delivering enantioenriched allyl phosphonates (**2i**–**2x**) with high yields and excellent chemo-, regio- and enantioselectivities. Heteroaryl substituents, such as 3-benzofuranyl and 3-benzothiophenyl, were also well tolerated, producing **2y** with 90% yield and 98% e.e., and **2z** with 50% yield and 86% e.e. Remarkably, biallenyl phosphonate proved to be a viable substrate, resulting in product **2aa** with superb selectivity.

Among the established methods for hydrogenating functionalized allenes, the presence of a single functional group often restricts the structural diversity of the resulting hydrogenation products[16,17]. By contrast, our cobalt catalytic system effectively accommodated a diverse array of functional groups, including phosphonates (**2ab** and **2ac**), phosphine oxide (**2ad**), sulfones (**2ae**–**2ag**) and esters (**2ah** and **2ai**), all of which yielded the desired products with satisfactory outcomes. Further extending the utility of this method, we explored the late-stage functionalization of biologically active molecules. Allenes derived from drug molecules or natural products were efficiently transformed into the desired products (**2aj**–**2an**) with high yields and excellent selectivities. Notably, derivatives from camphor sulfonic acid (**2aj**) and (−)-camphanic acid (**2ak**) underwent transformation without hydrogenation of the ketone and lactone units, showcasing the remarkable chemoselectivity of our system towards different types of unsaturated bond. Particularly noteworthy was the retention of the *Z*-configured double bond in the erucic acid derivative (**2am**), illustrating the catalyst's precise discrimination between different C=C double bonds. This selectivity ensured the targeted hydrogenation of allene occurred without the hydrogenation or isomerization of disubstituted olefins.

Next, we expanded the scope of cobalt-catalysed 2,3-hydrogenation of tetra-substituted allenes, as shown in Table 2. Tri-substituted alkenyl phosphonates, which are prevalent in functional molecules, serve as crucial precursors for various stereoselective transformations[57]. Thus, developing efficient methods to diversely synthesize both stereoisomers of trisubstituted alkenyl phosphonates from a single starting material is highly valuable. Under our optimized conditions, a broad array of tetra-substituted allenyl phosphonates underwent *E*-selective 2,3-hydrogenation using ligand **^{R}L12**, yielding the desired products in high yields with excellent stereo- and regioselectivities ((*E*)-**3a**–(*E*)-**3e**). Moreover, using ligand **^{R}L16** for the hydrogenation of these same allenes resulted in the corresponding (*Z*)-alkenyl phosphonates ((*Z*)-**3a**–(*Z*)-**3e**), with yields ranging from 75% to 92% and exceptional *Z*-selectivity between 10:1 and 99:1.

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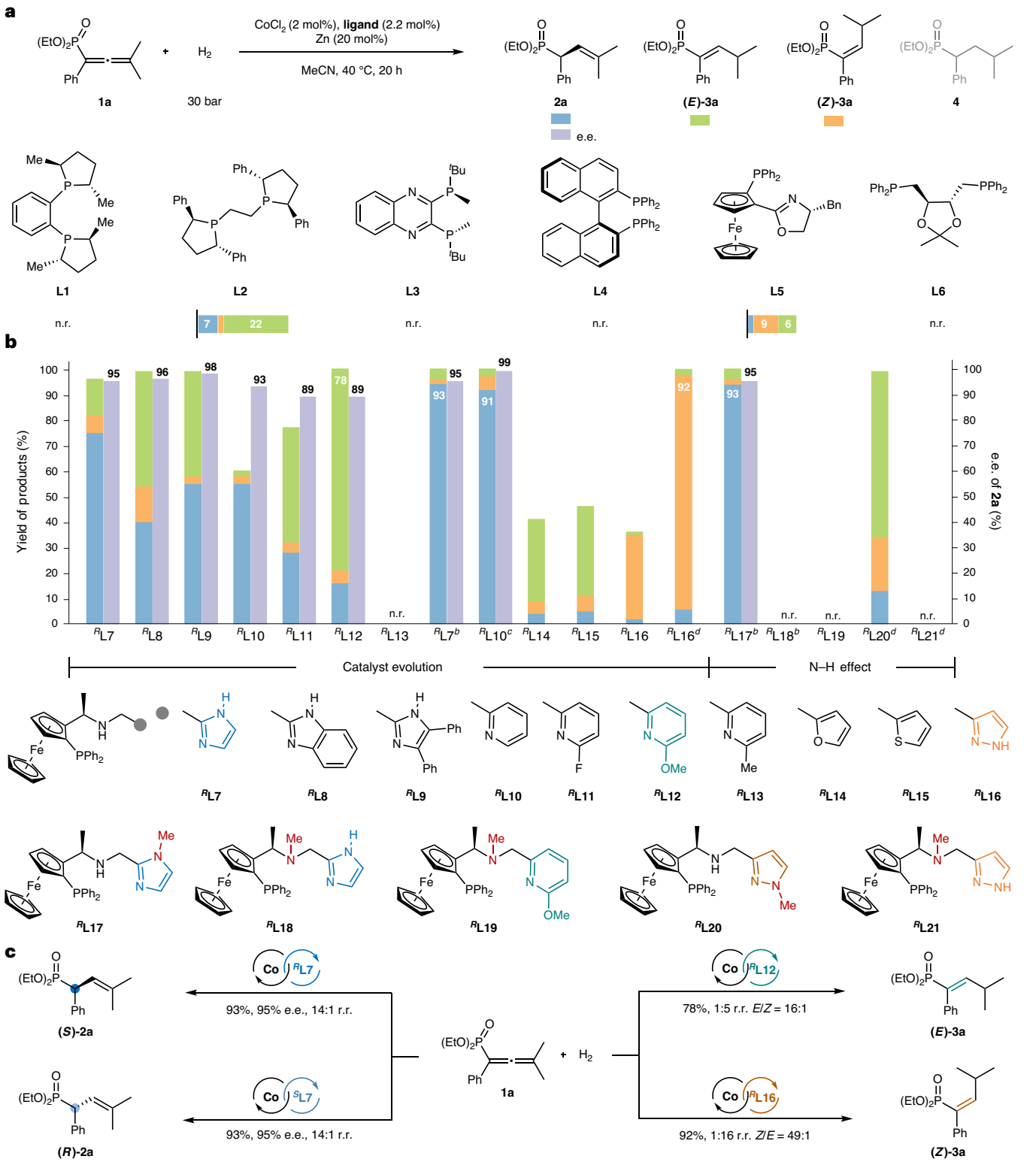
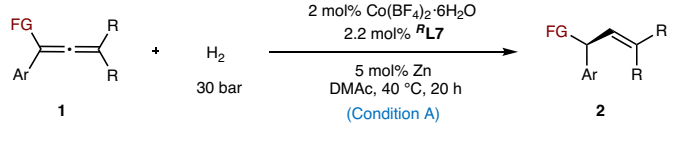
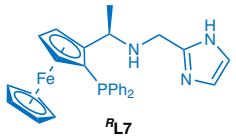


Fig. 2 | Overview of reaction development. a, Investigation of bidentate ligands. **b**, Effect of the pincer ligands on selectivity control and investigation of 'N–H' effect in pincer ligands. **c**, Divergent hydrogenation of tetra-substituted allenes to access all semihydrogenation products. [a]Reactions were performed on a 0.25 mmol scale using 2 mol% $CoCl_2$, 2.2 mol% ligand and 20 mol% Zn at 40 °C in 1 mL of MeCN for 20 h. Yields (%) and r.r. (**2**/**3**) were determined by gas chromatography (GC) analysis, and enantioselectivities (% e.e.) were determined by chiral-phase HPLC. [b]$Co(BF_4)_2{\cdot}6H_2O$ (2 mol%), ligand (2.2 mol%) and zinc dust (5 mol%) in 1 ml of DMAc. [c]$Co(OTf)_2$ (2 mol%), ligand (2.2 mol%) and zinc dust (20 mol%) in 1 ml of MeCN. [d]$CoCl_2$ (4 mol%), ligand (4.4 mol%) and zinc dust (40 mol%) in 1 ml of MeCN for 40 h. n.r., no reaction.

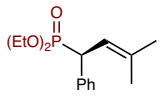
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Table 1 | Substrate scope of asymmetric 1,2-hydrogenation

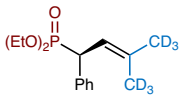




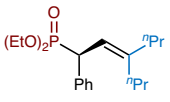
Variation of allene



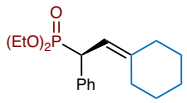
2a
93%, 95% e.e.
14:1 r.r.



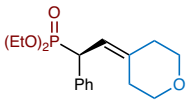
2b
91%, 95% e.e.
13:1 r.r.



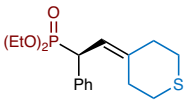
2c
87%, 96% e.e.
9:1 r.r.



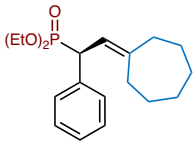
2d[b]
88%, 98% e.e.
9:1 r.r.



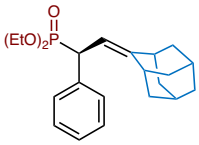
2e
86%, 96% e.e.
8:1 r.r.



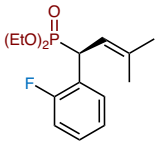
2f[b]
94%, 99% e.e.
22:1 r.r.



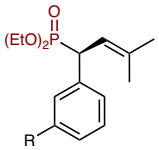
2g[b]
88%, 96% e.e.
8:1 r.r.



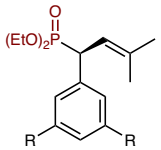
2h
95%, 94% e.e.
16:1 r.r.



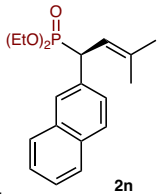
2i
86%, 93% e.e.
7:1 r.r.



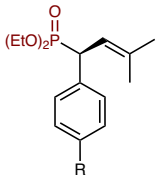
2j, R = Me, 90%, 94% e.e., 11:1 r.r.
2k, R = F, 90%, 95% e.e., 11:1 r.r.



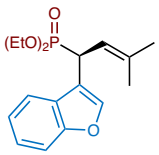
2l, R = OMe, 88%, 96% e.e., 10:1 r.r.
2m, R = F, 90%, 94% e.e., 12:1 r.r.



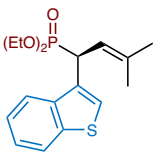
2n
90%, 93% e.e.
12:1 r.r.



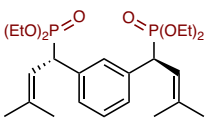
2o, R = Me, 92%, 96% e.e., 16:1 r.r.
2p, R = nPr, 88%, 95% e.e., 9:1 r.r.
2q, R = OMe, 91%, 96% e.e., 14:1 r.r.
2r, R = Ph, 85%, 95% e.e., 13:1 r.r.
2s, R = OH, 92%, 92% e.e., 17:1 r.r.
2t, R = CF_3, 90%, 97% e.e., 12:1 r.r.
2u, R = F, 92%, 96% e.e., 16:1 r.r.
2v, R = Cl, 88%, 92% e.e., 8:1 r.r.
2w, R = CO_2Me, 93%, 94% e.e., 21:1 r.r.
2x, R = CN, 87%, 95% e.e., 9:1 r.r.



2y
90%, 98% e.e.
14:1 r.r.

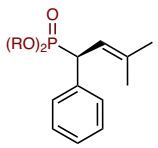


2z[b]
50%, 86% e.e.
5:1 r.r.

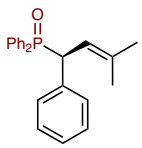


2aa[c]
92%, 99% e.e., > 20:1 d.r.
> 20:1 r.r.

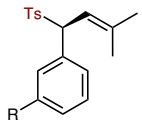
Variation of functional group



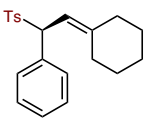
2ab, R = Me, 91%, 96% e.e., 13:1 r.r.
2ac, R = nBu, 91%, 95% e.e., 14:1 r.r.



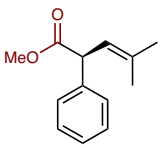
2ad
67%, 88% e.e.
3:1 r.r.



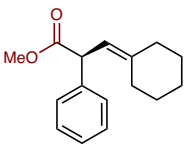
2ae[d], R = H, 64%, 90% e.e., 6:1 r.r.
2af[d], R = Me, 58%, 81% e.e., 8:1 r.r.



2ag[d]
43%, 96% e.e.
4:1 r.r.

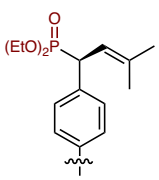


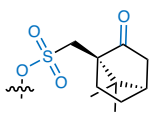
2ah[d]
75%, 70% e.e.
4:1 r.r.



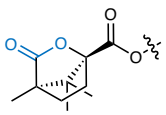
2ai[d]
80%, 80% e.e.
7:1 r.r.

Late-stage diversification

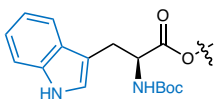




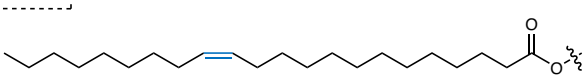
2aj[c]
from camphor sulfonic acid
90%, > 20:1 d.r., 11:1 r.r.



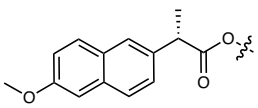
2ak[c]
from (–)-camphanic acid
88%, > 20:1 d.r., 9:1 r.r.



2al[c]
from tryptophan
87%, > 20:1 d.r., 9:1 r.r.



2am[c]
from erucic acid
91%, 94% e.e., 18:1 r.r.



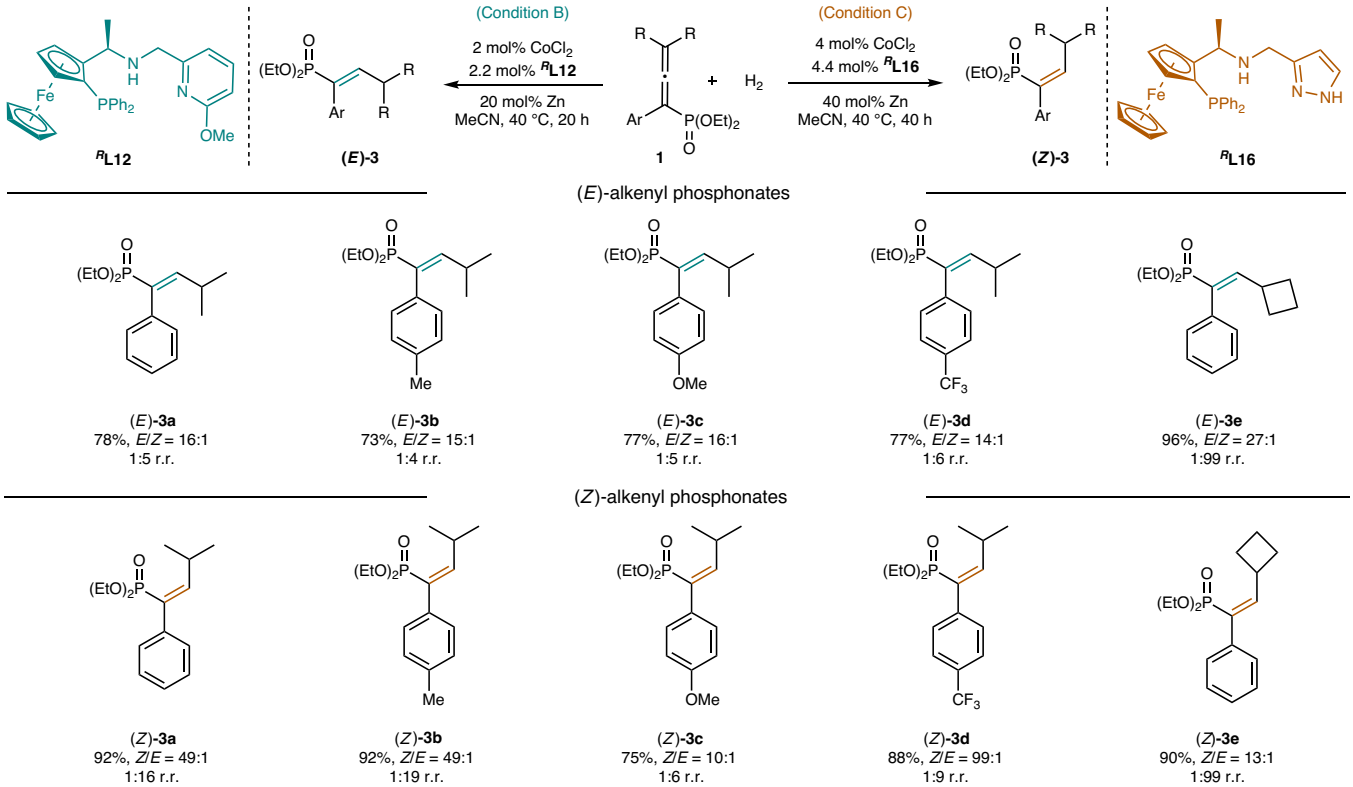
2an[c]
from naproxen
90%, > 20:1 d.r., 14:1 r.r.

[a]Reaction conditions: **1** (0.25 mmol), $Co(BF_4)_2{\cdot}6H_2O$ (2 mol%), **RL7** (2.2 mol%) and zinc dust (5 mol%) in 1 ml of DMAc at 40 °C for 20 h. Isolated yields (%) were given; r.r. (**2**/**3**) was determined by GC and e.e. (%) was determined by chiral-phase HPLC. [b]$Co(OTf)_2$ (2 mol%), **RL10** (2.2 mol%) and zinc dust (20 mol%) in 1 ml of MeCN at 40 °C for 20 h. [c]$Co(BF_4)_2{\cdot}6H_2O$ (4 mol%), **RL7** (4.4 mol%) and zinc dust (10 mol%), r.r. (**2**/**3**) and d.r. were determined by ^{1}H NMR. [d]$CoCl_2$ (4 mol%), **RL10** (4.4 mol%) and zinc dust (40 mol%) in 1 ml of MeCN at 40 °C for 40 h. FG=P(O)(OR)$_2$, P(O)Ph_2, Ts, CO_2Me.

Encouraged by the promising results from the non-chiral tetrasubstituted allenes, we next explored the divergent hydrogenation of racemic trisubstituted allenes, as shown in Extended Data Fig. 2. To our satisfaction, racemic allene **5a** underwent efficient cobalt-catalysed stereoconvergent 1,2-hydrogenation, producing the desired enantioenriched allyl phosphonate (*S*)-**6a** in high yields with excellent chemo-, regio- and stereoselectivities (78% yield, 97% e.e., >19:1 *E*/*Z*, and 4:1 r.r.). The opposite enantiomer of the hydrogenation product, (*R*)-**6a**,

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Table 2 | Substrate scope of 2,3-hydrogenation



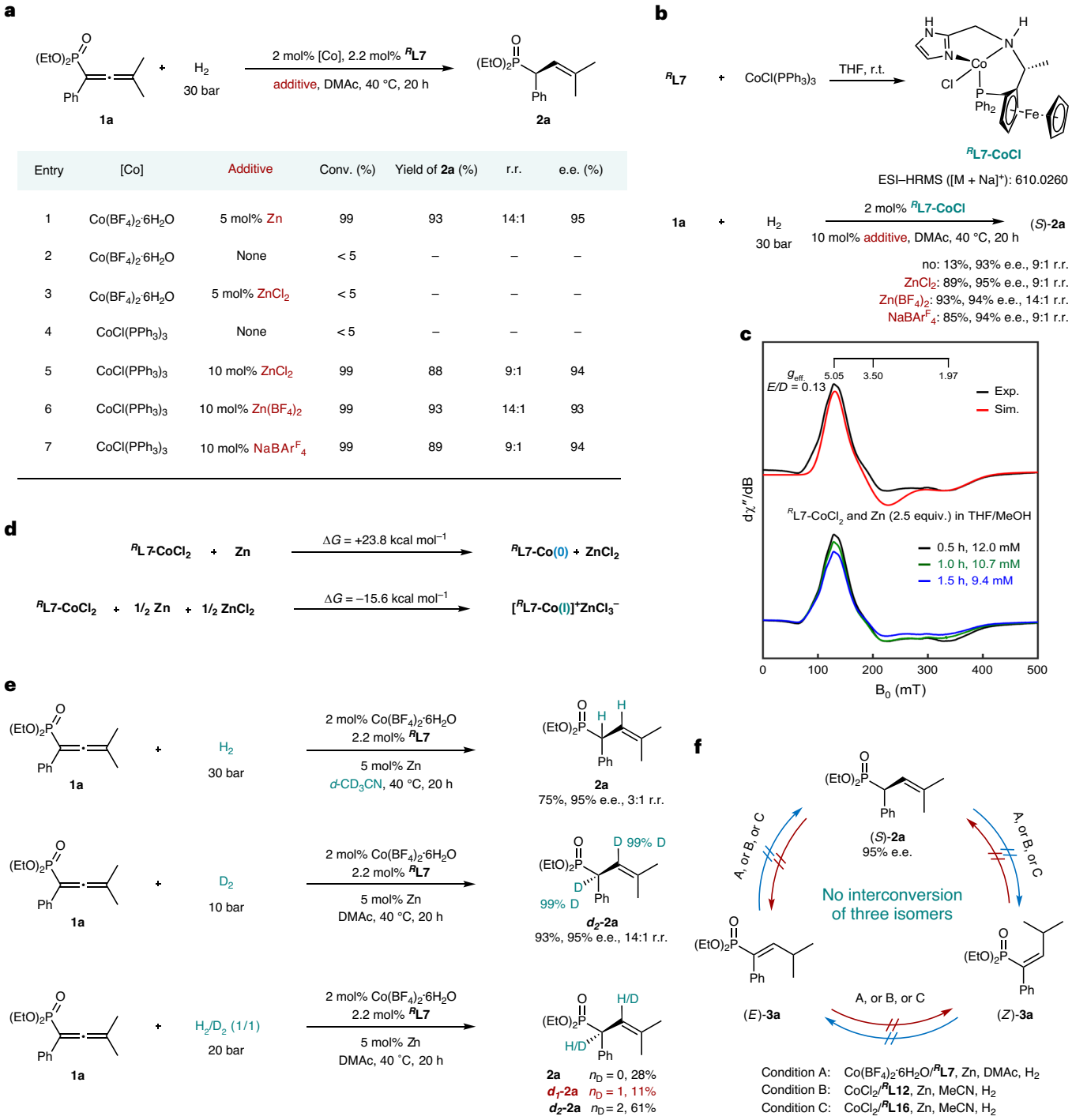
Reaction conditions for the synthesis of (*E*)-alkenyl phosphonates: **1** (0.25 mmol), $CoCl_2$ (2 mol%), **^{R}L12** (2.2 mol%) and zinc dust (20 mol%) in MeCN (1 ml) under H_2 (30 bar) at 40 °C for 20 h. Reaction conditions for the synthesis of (*Z*)-alkenyl phosphonates: **1** (0.25 mmol), $CoCl_2$ (4 mol%), **^{R}L16** (4.4 mol%) and zinc dust (40 mol%) in MeCN (1 ml) under H_2 (30 bar) at 40 °C for 40 h. Isolated yields (%) were given; r.r. (**2**/**3**) and *E*/*Z* ratio were determined by GC.

was obtained by using the enantiomer of optimized ligand (Extended Data Fig. 2a). This approach proved both robust and versatile, with the substitution pattern on the allene exerting minimal impact on reaction efficiency and selectivity profiles (Extended Data Fig. 2b). To gain preliminary insights into this stereoconvergent 1,2-hydrogenation process, we observed a notable erosion of the enantiomeric purity of both (−)-**5a** and (+)-**5a** under standard conditions without H_2 gas (Supplementary Fig. 32), indicating a racemization process. In addition, the hydrogenation of either (−)-**5a** or (+)-**5a** under standard conditions yielded the product (*S*)-**6a** with nearly identical enantioselectivity (Supplementary Fig. 33). These results suggest that our cobalt-catalysed stereoconvergent 1,2-hydrogenation of racemic allenes proceeds through a dynamic kinetic resolution mechanism. Notably, by using the optimized ligands for 2,3-hydrogenation of allenes, we also achieved stereodivergent 2,3-hydrogenation of **5a**, yielding the alkenyl phosphonates (*E*)-**7a** and (*Z*)-**7a** with high regioselectivity.

Mechanistic investigations

To gain insights into the mechanism of this cobalt-catalysed hydrogenation reaction, a set of mechanistic studies was performed (Fig. 3). Initial controlled experiments revealed that no product formed in the absence of a zinc reductant or with only the presence of $ZnCl_2$ (Fig. 3a, entries 2 and 3). Similarly, the catalyst formed in situ from $CoCl(PPh_3)_3$ and ligand **^{R}L7** failed to yield any product (Fig. 3a, entry 4). However, the addition of 10 mol% $ZnCl_2$, $Zn(BF_4)_2$ or $NaBAr^F{}_4$ dramatically enhanced the reaction efficiency, producing **2a** with high yields and enantioselectivity comparable to those achieved using the $Co(BF_4)_2{\cdot}6H_2O$/**^{R}L7**/Zn system (Fig. 3a, entries 5–7). Then, we synthesized **^{R}L7-CoCl** complex via treating $CoCl(PPh_3)_3$ with ligand **^{R}L7** (Fig. 3b). This isolated Co(I) complex exhibited catalytic activity in the hydrogenation of **1a** under zinc-free conditions, with additives such as $ZnCl_2$, $Zn(BF_4)_2$ and $NaBAr^F{}_4$, producing **2a** with high yields and enantioselectivity comparable to those achieved using CoX_2/**^{R}L7**/Zn system. This suggests that the reaction may be initiated by an active cationic Co(I) species[6,30,34]. Further mechanistic insight was gained from in situ electron paramagnetic resonance (EPR) experiments (Fig. 3c). We observed a decrease in the signal of Co(II) complex[58], **^{R}L7-CoCl$_2$**, and no evidence of Co(0) species was found[29], indicating a single-electron reduction of the Co(II) complex to Co(I) species using zinc dust within 1.5 h. The in situ high-resolution mass spectrometry analysis of this reduction process also confirmed the generation of Co(I) species, with the detection of **^{R}L7-CoCl** (*m*/*z* = 610.0268, $[M + Na]^+$) (Supplementary Figs. 16 and 17). Density functional theory (DFT) calculations align with these experimental observations (Fig. 3d and Supplementary Fig. 36). The sequential reduction of the complex **^{R}L7-CoCl$_2$** by zinc to **^{R}L7-Co**(0) was calculated to be strongly endergonic (+23.8 kcal mol^{-1}), indicating a thermodynamically unfavourable process. By contrast, the reduction of **^{R}L7-CoCl$_2$** to generate the cationic **[^{R}L7-Co**(I)**]$^+$ZnCl$_3^-$** species, was found to be highly exergonic (−15.6 kcal mol^{-1}), suggesting that the formation of the catalytically active Co(I) species is thermodynamically favourable. These results support the hypothesis that zinc reduces the Co(II) precatalyst to a Co(I) complex, which is then activated by halide abstraction from ZnX_2 to form a catalytically active cationic Co(I) species. We also considered the Co(II)–H mechanism, where the N–H group was deprotonated to give an amido group that can also facilitate metal–ligand cooperative heterolytic cleavage of H_2 with cobalt centre[48,51]. When various bases were used instead of reductants to activate the cobalt(II) catalyst precursor, no reaction occurred, indicating the amino Co(II)–H complex is not the major catalytically active species in our catalytic system (Supplementary Table 8). In the deuterium labelling experiments, hydrogenation

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Entry	[Co]	Additive	Conv. (%)	Yield of **2a** (%)	r.r.	e.e. (%)
1	$Co(BF_4)_2{\cdot}6H_2O$	5 mol% Zn	99	93	14:1	95
2	$Co(BF_4)_2{\cdot}6H_2O$	None	< 5	–	–	–
3	$Co(BF_4)_2{\cdot}6H_2O$	5 mol% $ZnCl_2$	< 5	–	–	–
4	$CoCl(PPh_3)_3$	None	< 5	–	–	–
5	$CoCl(PPh_3)_3$	10 mol% $ZnCl_2$	99	88	9:1	94
6	$CoCl(PPh_3)_3$	10 mol% $Zn(BF_4)_2$	99	93	14:1	93
7	$CoCl(PPh_3)_3$	10 mol% $NaBAr^F{}_4$	99	89	9:1	94

Fig. 3 | Mechanistic investigations. a, Controlled experiments. These results indicate that the catalytically active species in this reaction may be a cationic Co(I) species, rather than a Co(II) species. **b**, Synthesis and evaluation of ^{R}**L7-CoCl** complex. **c**, EPR experiments. All spectra were recorded at 10 K. The Co(II) ($S = 3/2$) spectrum was simulated by using a **g** tensor of [2.17, 2.15, 2.06]; $A(^{59}Co) = [200, 30, 30]$ MHz; $g_{strain} = [0.7, 0.6, 0.5]$; $E/D = 0.13$, and $D \gg 0.8$ cm^{-1}. **d**, Calculated Gibbs free energy for the preactivation process of precatalyst Co^{II} to Co^0 and Co^I by Zn atom and $ZnCl_2$. Energy values are in kcal mol^{-1}, and Gibbs free energy was calculated at the M06L/Def2-TZVP/SMD(DMAc)//M06L/Def2-SVP level of theory. **e**, Deuterium labelling experiments. The quantitative mass spectrum was determined on the basis of isotopic distributions. **f**, Study on the possibility of isomerization of three products. HRMS, high-resolution mass spectrometry; r.t., room temperature. Conv., conversion; B_0, magnetic field; Exp., experimental spectrum; Sim., simulated spectrum; D and E are zero-field splitting paramaters, with D describing the axial component of the magnetic dipole–dipole interaction and E as the transversal component.

of **1a** using deuterium gas (D_2) yielded the desired product ***d*$_2$-2a**, with 99% deuterium incorporation (Fig. 3e). When the reaction was conducted with hydrogen gas in CD_3CN, no deuteration of the product was detected. These findings confirm that H_2 serves as the exclusive hydrogen source in this reaction. In addition, mass spectrometry and NMR characterization of the product obtained under mixed gas conditions ($H_2/D_2 = 1/1$) revealed the presence of a mono-deuterated product, ***d*$_1$-2a** (Supplementary Figs. 20–22). This suggests that the two hydrogen atoms added to the product may come from different H_2 molecules. Further tests were conducted by subjecting three

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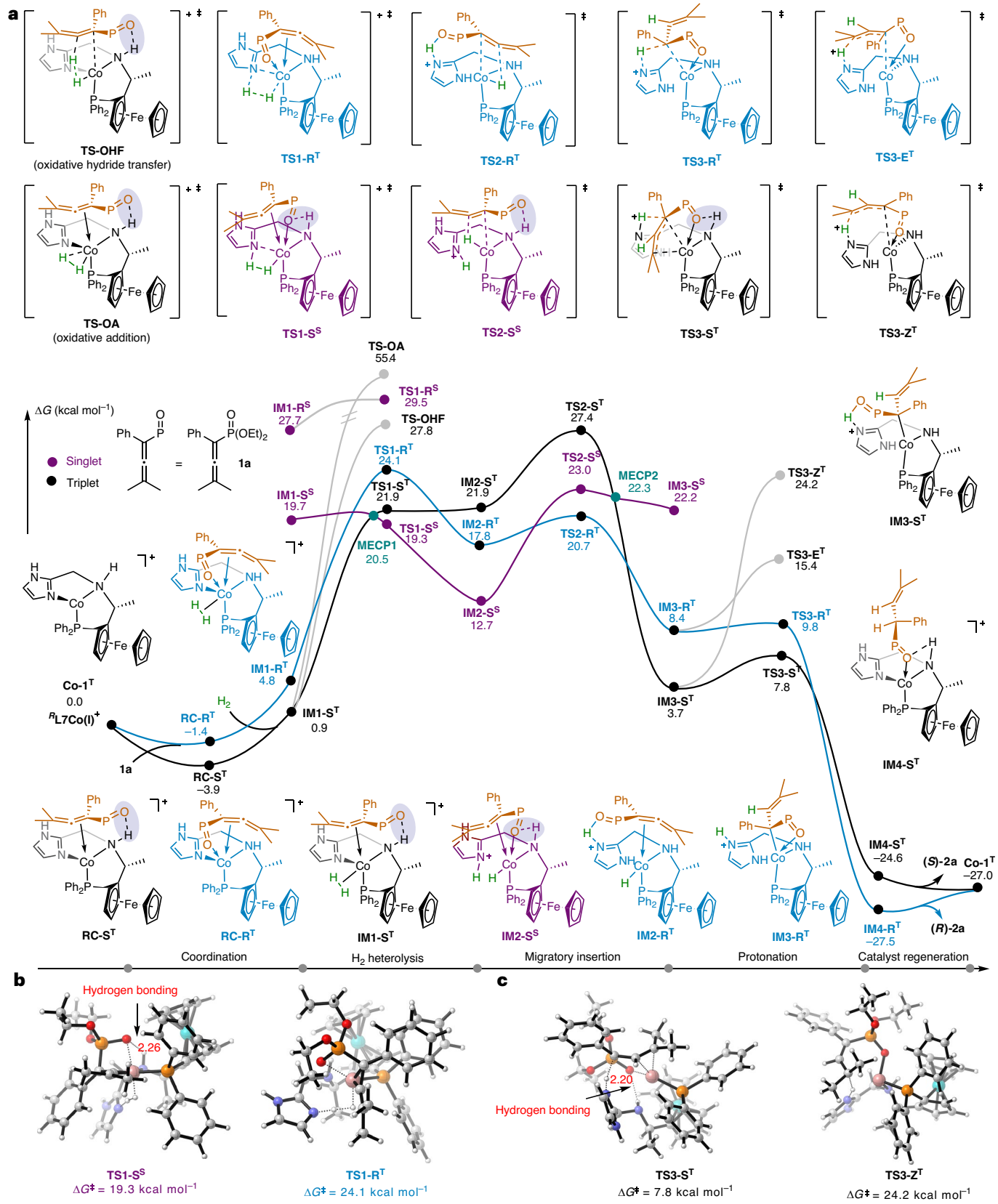


Fig. 4 | Computational investigations. a, Gibbs free energy profile of redox neutral mechanism. The superscripts S and T are used to denote the singlet and triplet states of the calculated species. ΔG values (in kcal mol^{-1}) were calculated at the M06L/Def2-TZVP/SMD(DMAc)//M06L/Def2-SVP level of theory. **b**, Optimized transition states of **TS1-S^S** and **TS1-R^T**. For H_2 heterolysis transition states, **TS1-S^S** is favoured over **TS1-R^T** by 4.8 kcal mol^{-1}. **c**, Optimized transition states of **TS3-S^T** and **TS3-Z^T**. For protonation transition states, **TS3-S^T** is favoured over **TS3-Z^T** by 16.4 kcal mol^{-1}. MECP, minimum energy crossing point.

semihydrogenation products, **2a**, (*E*)-**3a** and (*Z*)-**3a**, to their respective optimized reaction conditions (Fig. 3f). No conversion or isomerization events were observed, confirming that the product distribution is solely controlled by the catalysts in hydrogenation process across different reaction pathways.

Based on the experimental results, three potential pathways were proposed and calculated: redox-active pathways involving oxidative hydride transfer and oxidative addition-reductive elimination mechanisms, as well as a redox-neutral mechanism involving H_2 heterolysis (Extended Data Fig. 3). The simplified Gibbs free energy profile for the kinetically favoured redox-neutral pathway is presented in Fig. 4, while detailed elementary steps for the redox-active pathways are provided in Supplementary Figs. 37–39. With Co(I) cation species ([^{R}**L7-Co**]$^+$) as the starting point, coordination of **1a** to the Co centre via its *Si* or *Re* face forms intermediates **RC-S^T** and **RC-R^T**, respectively. Notably, the formation of **RC-S^T** is more favourable due to the hydrogen-bonding interaction between the phosphonate group and catalyst. In addition, the coordination of **1a** with cobalt centre in the absence of hydrogen-bonding interaction (**RC-S$^T_{nm}$**), as well as from opposing face of catalyst with hydrogen-bonding interaction (**RC-S^T-NH$_{anti}$**) were also investigated (Supplementary Figs. 40–42). It was found that **RC-S$^T_{nm}$** and **RC-S^T-NH$_{anti}$** are less favoured by 3.7 and 4.0 kcal mol^{-1}, respectively. Subsequent coordination of H_2 leads to the formation of **IM1-S^T** and **IM1-R^T**, which are endergonic by 4.8 and 8.7 kcal mol^{-1}. Following this, the imidazole group and cobalt centre undergo a metal–ligand cooperative H_2 heterolytic cleavage via transition states **TS1-S^S** and **TS1-R^T**, generating Co–H species **IM2-S^S** and **IM2-R^T** bearing a protonated imidazole group, with free energy barriers of 23.2 and 28.0 kcal mol^{-1}, respectively. Interestingly, in this step, the high-spin **IM1-S^T** can undergo a spin-crossover process via a minimum energy crossing point (**MECP1**), transforming into the low-spin intermediate **IM1-S^S**. By contrast, this spin-crossover does not occur in the H_2 cleavage of **IM2-R^T** via transition state **TS1-R^T**. This difference arises from the distinct coordination modes, which depend on the presence or absence of hydrogen-bonding interaction, underscoring the critical role of MLC (Fig. 4b). Moreover, the redox-active pathways involving oxidative addition and oxidative hydride transfer, as represented by transition states **TS-OA** and **TS-OHF**, were found to be 36.1 and 8.5 kcal mol^{-1} less favourable, respectively, compared with **TS1-S^S**. This is further supported by the presence of the mono-deuterated product ***d$_1$*-2a** in deuterium labelling experiments. The allene insertion into Co–H species occurs to generate the allyl cobalt intermediate **IM3-S^S** via **TS2-S^S** with an energy barrier of 10.3 kcal mol^{-1}. The activation strain model (ASM) analysis[59] reveals that, although rate-determining **TS2-S^S** exhibits higher distortion energy than the hydride transfer transition state **TS-OHF** in the redox-active pathway (Supplementary Fig. 43), it benefits from stronger interaction energy, which leads to a lower overall energy barrier for the stepwise reaction pathway involving H_2 cleavage and allene insertion. **IM3-S^S** then undergoes a spin-crossover via **MECP2**, converting into the more stable triplet **IM3-S^T**. The frontier molecular orbital and energy decomposition analyses (EDA)[60] results reveal critical electronic structures driving the triplet-singlet transitions (Supplementary Figs. 44 and 45). The antibonding character of singly occupied molecular orbitals (SOMOs) destabilizes triplet transition states of **TS1-S^T** and **TS2-S^T**, whereas the bonding characters and lower HOMO energy in singlet transition states of **TS1-S^S** and **TS2-S^S** enhance stability. Furthermore, the EDA results highlight that the singlet states of **TS1-S^S** and **TS2-S^S** are stabilized by stronger orbital interactions with bonding character. Meanwhile, the transformation of **IM2-R^T** into **IM3-R^T** through **TS2-R^T** has a lower energy barrier of 2.9 kcal mol^{-1}, maintaining the triplet state. This reduction in the energy barrier can be attributed to the smaller distortion and formation of new hydrogen-bonding interaction between the phosphonate group and the protonated imidazole, which stabilizes the transition state (Supplementary Fig. 46). The final step involves proton transfer from the imidazole moiety, leading to the formation of (*S*)-**2a** and (*Z*)-**3a** from **IM3-S^T** via transition states **TS3-S^T** and **TS3-Z^T**, as well as (*R*)-**2a** and (*E*)-**3a** from **IM3-R^T** through transition states **TS3-R^T** and **TS3-E^T**, respectively. Notably, the 1,2-hydrogenation is more favourable than 2,3-hydrogenation as a result of hydrogen-bonding interactions that serve to stabilize the transition state **TS3-S^T** (Fig. 4c). Analysing the entire Gibbs energy profile, the overall energy barriers for forming (*S*)-**2a**, (*R*)-**2a**, (*Z*)-**3a** and (*E*)-**3a** are 26.9, 28.0, 28.1 and 28.0 kcal mol^{-1}, with **TS2-S^S**, **TS1-R^T**, **TS3-Z^T** and **TS1-R^T** identified as the rate-determining steps, respectively. Consequently, the formation of (*S*)-**2a** is the most favoured, which aligns well with the experimentally observed enantio- and regioselectivity. To further understand the origin of enantiocontrol favouring the formation of (*S*)-**2a**, we conducted an independent gradient model based on Hirshfeld partition (IGMH) analysis[61] on the transition states involved in the enantioselectivity-determining step (Supplementary Fig. 47). The IGMH analysis revealed a strong N–H···O interaction between the catalyst and the substrate in **TS1-S^S** (Fig. 4b). ASM analysis further supported this observation, showing a significantly stronger interaction energy in **TS1-S^S** (−32.3 kcal mol^{-1}) compared with **TS1-R^T** (−16.4 kcal mol^{-1}) (Supplementary Fig. 48). The EDA result indicates that enhanced exchange–correlation energy and orbital interaction energy in **TS1-S^S** contribute to the selectivity (Supplementary Fig. 49). Overall, the N–H···O hydrogen-bonding interaction is pivotal, not only by liberating a coordination site for H_2 activation but also by effectively differentiating the prochiral faces of the alkene, thus playing a crucial role in determining the reactivity and enantioselective outcome of this transformation.

Synthetic applications

Deuterium-incorporated compounds are of important interest due to their broad utility across various fields, particularly in drug discovery and development[62]. However, achieving selective deuterium incorporation at specific sites with high isotope enrichment remains a challenging task. We are pleased to report the successful synthesis of a series of deuterated products, including ***d$_2$*-2a**, **2e**, **2o**, **2ac**, (*E*)-**3a** and (*Z*)-**3a**, under optimized reaction conditions using 10 bar of D_2. These products were obtained with up to 99% deuterium incorporation at both sp^2 and sp^3 carbons (Extended Data Fig. 4a). Achieving such high levels of deuterium incorporation at these two positions is difficult with conventional methods. To illustrate the synthetic applicability of our approach, we successfully performed a gram-scale cobalt-catalysed 1,2-hydrogenation of **1a** with a low substrate/catalyst ratio of 500, yielding product **2a** in 91% with 95% e.e. (Extended Data Fig. 4b). Enantioenriched allylic phosphonates, important precursors for natural products and biologically active molecules, were further transformed into a variety of derivatives. As shown in Extended Data Fig. 4c, **2a** was converted into thiophosphonate **8**, phosphonic acid **9**, an epoxide compound **10**, and β-hydroxyl-phosphonate **13**, all in good-to-high yields while retaining enantiopurity. Furthermore, treatment of **2a** with NBS or NIS produced enantiopure 4-bromo and 4-iodo-cyclic phosphonates (**11** and **12**), bearing a P-chiral centre. These compounds are rare in the literature and difficult to synthesize by other methods. The structure and absolute configuration of **11** were confirmed by X-ray crystallography. Moreover, allenyl phosphonate **15**, synthesized from L-4-iodophenylalanine **14** in two steps, can be efficiently transformed into enantioenriched phosphonate **16** through our cobalt-catalysed asymmetric hydrogenation (Extended Data Fig. 4d). This reaction achieves a 90% yield with an impressive >20:1 diastereoselective ratio (d.r.) and 12:1 r.r., effectively producing a phosphorylated amino acids analogue[63], showcasing the versatility and effectiveness of our method in synthesizing complex molecular structures.

Conclusions

We have developed a highly divergent hydrogenation strategy for multisubstituted allenes, enabling the synthesis of a wide range of

stereodefined trisubstituted alkenes with excellent functional group tolerance and outstanding selectivity control. The success of this approach hinges on the use of *NNP*-pincer cobalt catalysts, which leverage multiple MLC functionalities, validated by detailed mechanistic studies. This work not only addresses a persistent challenge in allene chemistry but also introduces a general strategy to overcome the issue of insufficient coordination sites in transition-metal catalysis, paving the way for future advancements in selective and efficient catalytic transformations.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41557-025-01945-2.

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Methods

In an argon-filled glovebox, CoX_2 (2–4 mol%), ligand (^{R}**L7**, ^{R}**L12** or ^{R}**L16**, 2.2–4.4 mol%), zinc dust (5–40 mol%), substrate **1** (0.25 mmol) and 1.0 ml of solvent were added to a hydrogenation tube equipped with a stirring bar. The tube was then sealed and placed inside an autoclave. The autoclave was pressurized to 30 bar with hydrogen, and the reaction mixture was stirred at 40 °C for 20 or 40 h. Upon completion, the reaction mixture was purified using preparative thin-layer chromatography to yield the desired product. For additional experimental details, please refer to the Supplementary Information.

Data availability

The data that support the findings of this study are available in this Article and its Supplementary Information. The crystallographic data for the structures reported in this Article have been deposited at the Cambridge Crystallographic Data Centre under deposition numbers CCDC 2345206 (**11**), 2345208 (^{R}**L7-CoCl$_2$**), 2397603 (^{R}**L11-CoCl$_2$**), 2345207 (^{R}**L12-CoCl$_2$**), 2397604 (^{R}**L14-CoCl$_2$**), 2397605 (^{R}**L15-CoCl$_2$**) and 2397606 (^{R}**L17-CoCl$_2$**). Copies of the data can be obtained free of charge via https://www.ccdc.cam.ac.uk/structures.

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Author contributions

Conceptualization: X.R., S.M. and Q.L. Methodology: X.R. and Y.R. Investigation: X.R., Y.R., Y.C., C.L., J.W., Z.C. and L.T. Funding acquisition: Z.W. and Q. L. Project administration: L.T., S.M., Z.W. and Q.L. Supervision: L.T., S.M., Z.W. and Q.L. Writing—original draft: X.R., Z.W. and Q.L. Writing—review and editing: X.R. and Q.L.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41557-025-01945-2.

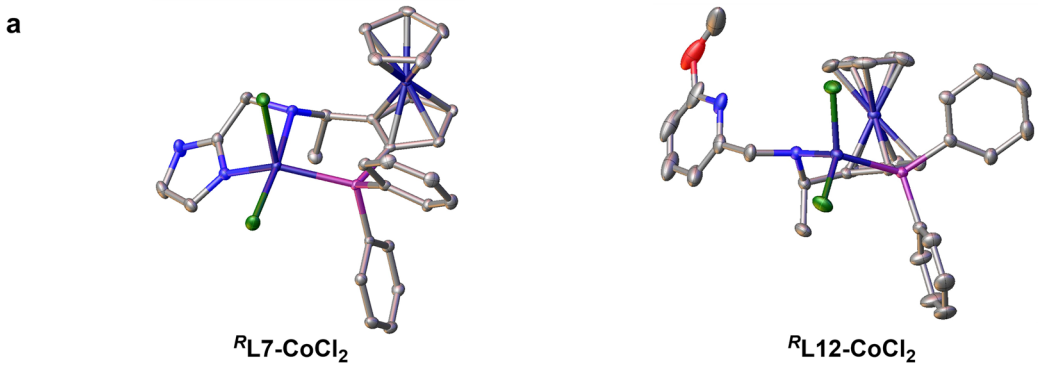
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41557-025-01945-2.

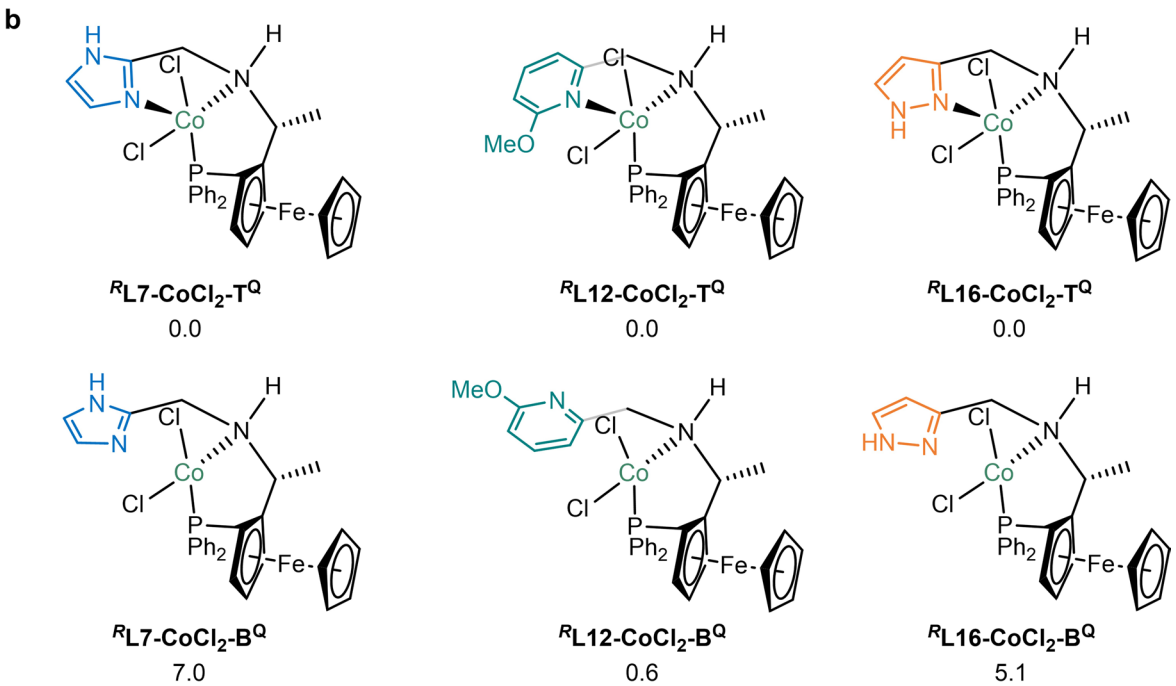
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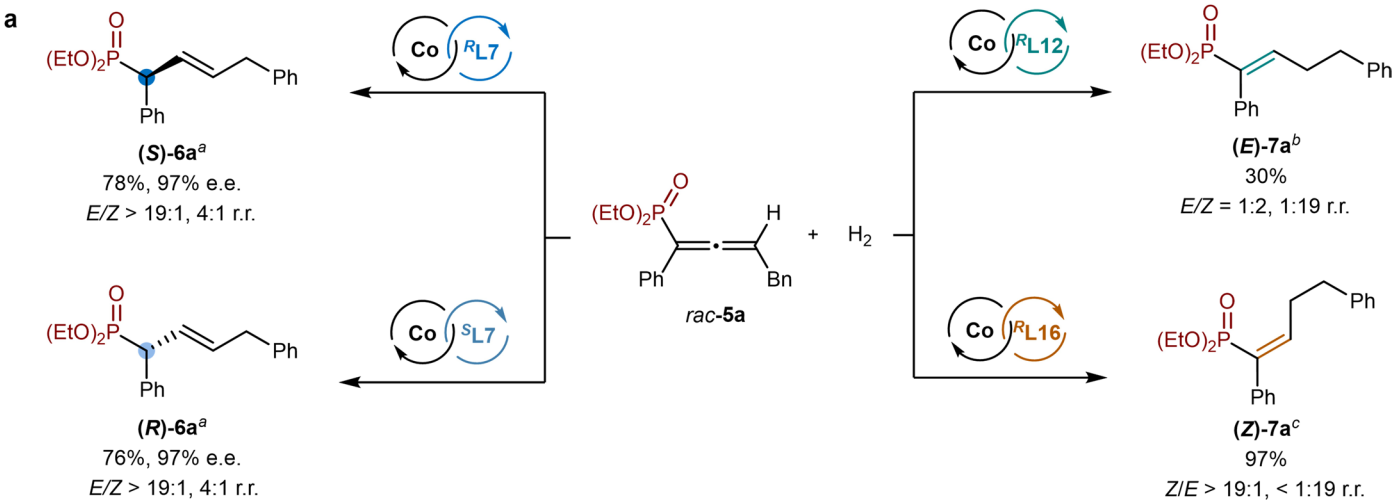
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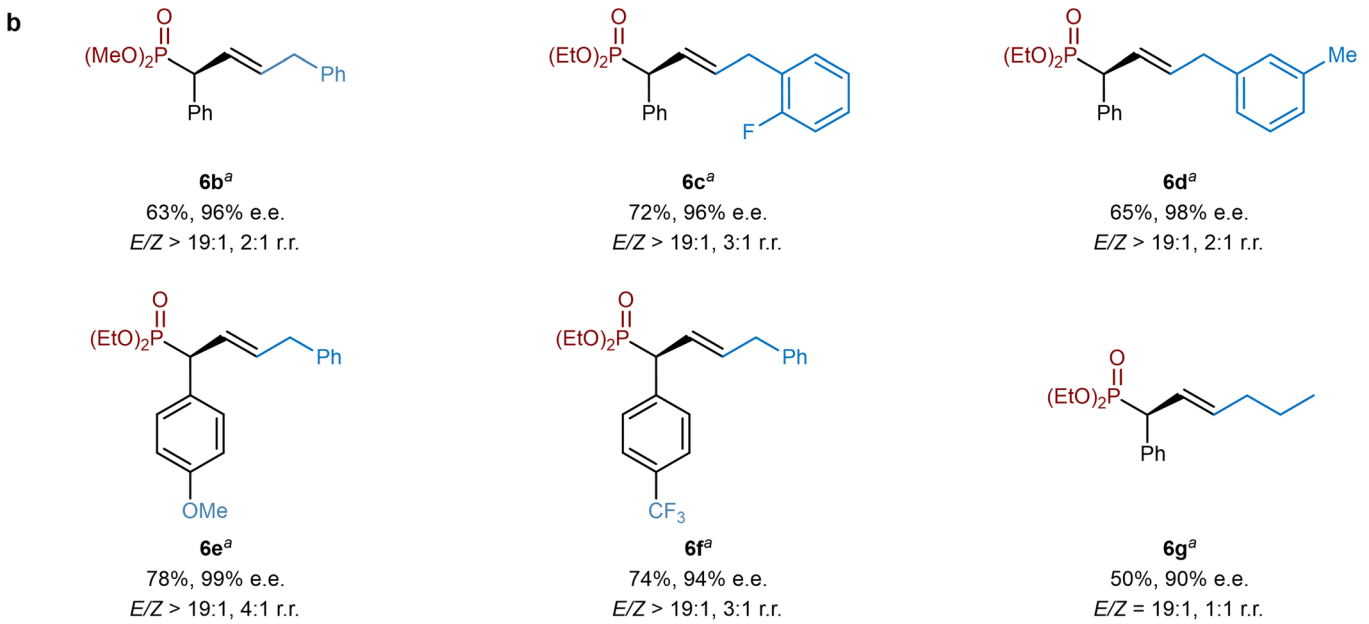




Extended Data Fig. 1 | Structure of cobalt catalysts. a, Solid-state structure of ^{R}**L7-CoCl$_2$** and ^{R}**L12-CoCl$_2$** determined by single-crystal X-ray diffraction. Ellipsoids set at 50% probability. Hydrogen atoms are omitted for clarity. Colour code for the atoms in the structures: grey, carbon; light blue, nitrogen; red, oxygen; purple, phosphorus; green, chlorine; royal blue, iron; dark blue, cobalt. **b**, Dissociation free energy of the N-heterocyclic ligand. T and B denote tridentate and bidentate coordination structures, respectively, and superscript Q indicates quartet spin states. ΔG values (in kcal mol^{-1}) were calculated at the M06L/def2-TZVP/SMD(DMAc)//M06L/def2-SVP level.

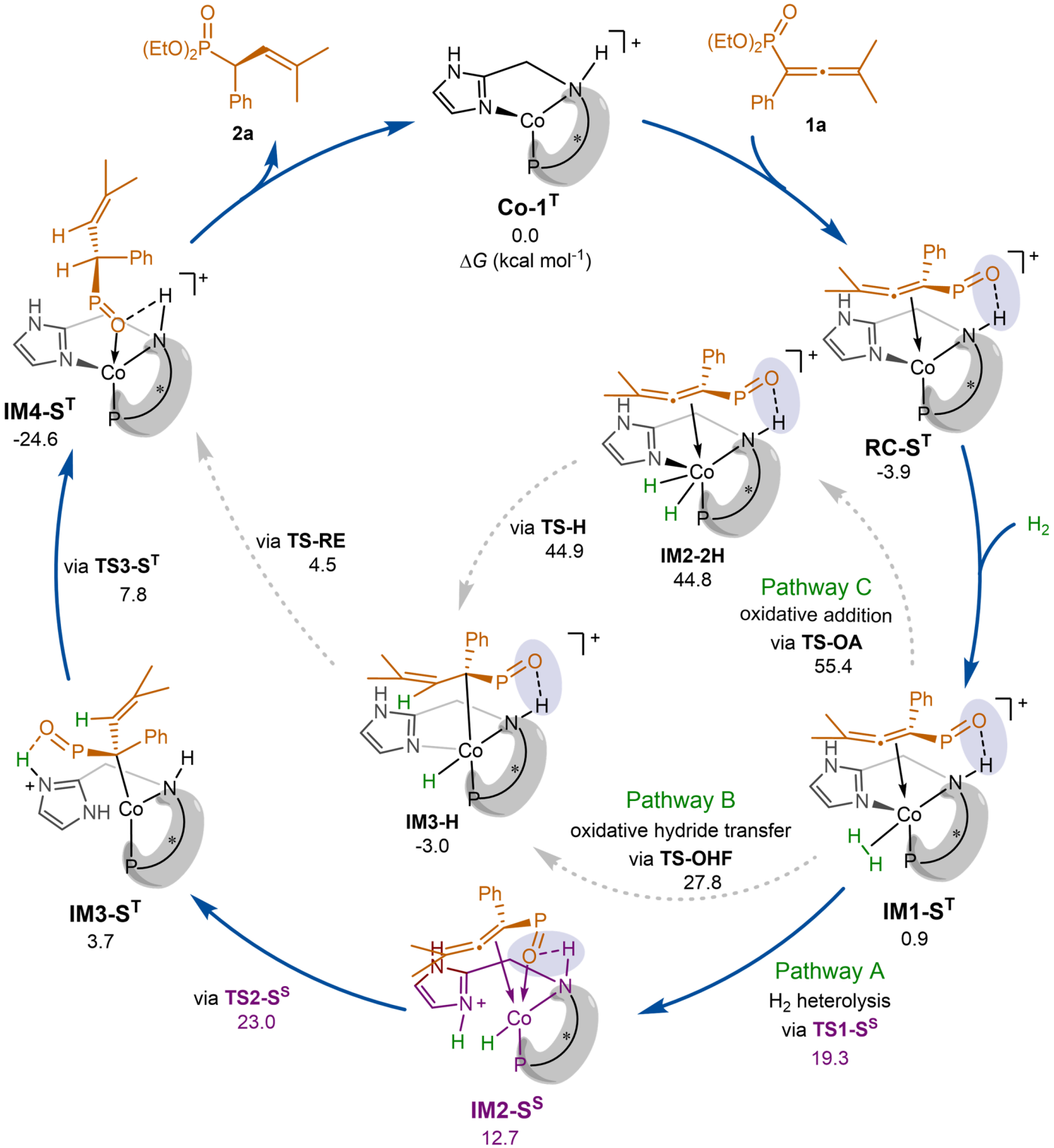
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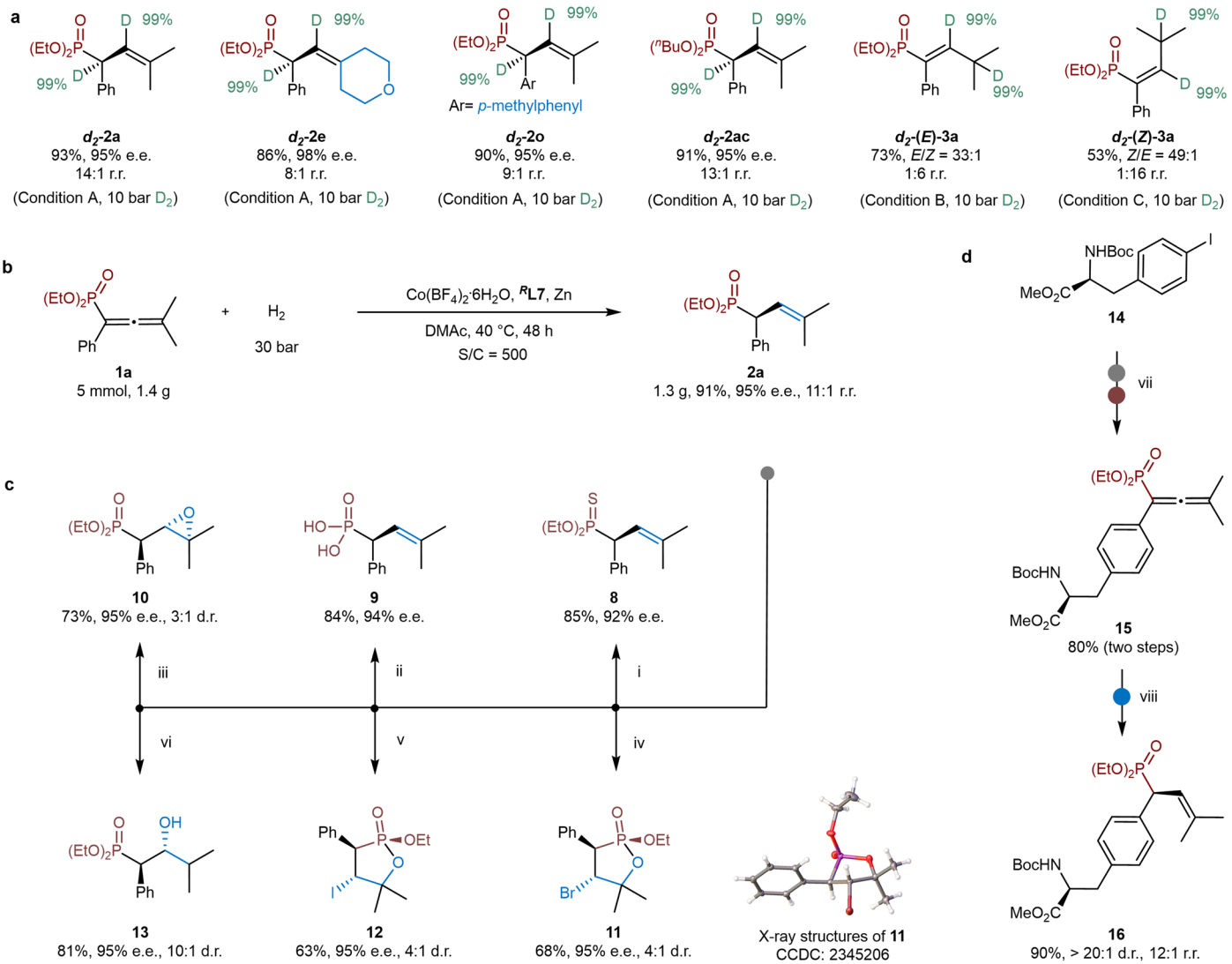
Extended Data Fig. 2 | Tunable cobalt-catalyzed hydrogenation of racemic trisubstituted allenes. a, Divergent hydrogenation of racemic trisubstituted allenes. **b**, Substrate scope of asymmetric 1,2-hydrogenation of racemic allenes. Reaction conditions: [a] *rac*-**5** (0.25 mmol), $Co(BF_4)_2{\cdot}6H_2O$ (2 mol%), ^{R}**L7** or ^{S}**L7** (2.2 mol%) and zinc dust (20 mol%) in 1 mL of DMAc at 40 °C for 20 h. Isolated yields (%) were given, regioselective ratio (r.r. **6**/**7**) was determined by ^{31}P NMR and enantiomeric excess (% e.e.) was determined by chiral-phase HPLC. [b] $CoCl_2$ (2 mol%), ^{R}**L12** (2.2 mol%) and zinc dust (20 mol%) in HFIP (1 mL) at 60 °C for 20 h. *E*/*Z* ratio was determined by ^{31}P NMR. [c] $CoCl_2$ (2 mol%), ^{R}**L16** (2.2 mol%) and zinc dust (20 mol%) in DMAc (1 mL) at 60 °C for 40 h. *Z*/*E* ratio was determined by ^{31}P NMR.

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Extended Data Fig. 3 | Catalytic cycle. Proposed mechanism for the stereoselective cobalt-catalyzed hydrogenation of functionalized allenes. DFT calculations were performed at the M06L/Def2-TZVP/SMD(DMAc)//M06L/Def2-SVP level of theory.

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Extended Data Fig. 4 | Synthetic Applications. a, Syntheses of deuterated trisubstituted alkene products. **b**, Gram-scale synthesis with 0.2 mol% catalyst. **c**, Post functionalisation of reaction products. **d**, Preparation of intermediate of phosphonomethyl-phenylalanine analogue. [i] Lawesson's reagent, toluene, 100 °C. [ii] TMSBr, DCM, r.t. [iii] *m*CPBA, −20 °C. [iv] NBS, DMSO/H_2O (v/v = 7/5), r.t. [v] NIS, DMSO/H_2O (v/v = 7/5), r.t. [vi] 1) BH_3, THF, -20 °C, 2) H_2O_2, NaOAc, 0 °C. [viii] 1) $Pd(PPh_3)_4$ (2 mol%), CuI (2 mol%), 3-Methyl butynol and Et_3N, 80 °C. 2) $P(OEt)_2Cl$, Et_3N, DCM, −78 °C - r.t. [viii] $Co(BF_4)_2·6H_2O$ (2 mol%), **RL7** (2.2 mol%) and zinc dust (5 mol%) in 1 mL of DMAc under H_2 (30 bar) at 40 °C for 20 h.